

ATLAS

Personal Life Operating System

Requirements Document -- v0.1.0

Repository: <https://github.com/Swayamjain1240/Atlas>

Table of Contents

- 1. Functional Requirements
- 2. Non-Functional Requirements
- 3. Security Requirements
- 4. UI/UX Requirements
- 5. Data Requirements
- 6. Integration Requirements
- 7. Performance Requirements
- 8. Deployment Requirements

1. Functional Requirements

1.1 Query System

- [FR-001]

Priority: CRITICAL

User shall be able to ask natural language questions about their indexed data
- [FR-002]

Priority: CRITICAL

System shall return answers with source citations and relevance scores
- [FR-003]

Priority: HIGH

System shall support multi-turn conversations with context retention
- [FR-004]

Priority: HIGH

User shall be able to filter queries by source type (files, browser, etc.)
- [FR-005]

Priority: MEDIUM

User shall be able to search directly by semantic similarity without LLM generation

1.2 Ingestion System

- [FR-010]

Priority: CRITICAL

System shall ingest text files (.txt, .md, .py, .js, .json, .csv, .html, .css, .yaml)
- [FR-011]

Priority: HIGH

System shall ingest PDF documents using text extraction
- [FR-012]

Priority: HIGH

System shall ingest Word documents (.docx)
- [FR-013]

Priority: HIGH

System shall ingest Chrome-based browser history (Chrome, Edge, Brave)
- [FR-014]

Priority: CRITICAL

System shall automatically chunk documents into searchable segments
- [FR-015]

Priority: MEDIUM

System shall support configurable watch directories
- [FR-016]

Priority: MEDIUM

System shall detect and ingest new/changed files on configured interval

1.3 Memory Management

- [FR-020]

Priority: HIGH

ATLAS -- Requirements Document

User shall be able to view recently indexed memories

[FR-021]

Priority: HIGH

User shall be able to delete specific memories

[FR-022]

Priority: MEDIUM

System shall display total memory statistics (count, sources, sizes)

[FR-023]

Priority: MEDIUM

User shall be able to trigger manual re-ingestion

2. Non-Functional Requirements

[NFR-001]

Priority: CRITICAL

All data processing must happen locally on the user's machine. Zero cloud dependency.

[NFR-002]

Priority: CRITICAL

Query response time shall be < 2 seconds for 95% of requests

[NFR-003]

Priority: HIGH

System shall operate fully offline without internet connection

[NFR-004]

Priority: HIGH

Vector search shall return results within 200ms for up to 100,000 documents

[NFR-005]

Priority: MEDIUM

System shall use less than 500MB RAM when idle

[NFR-006]

Priority: HIGH

System shall support Windows, macOS, and Linux

[NFR-007]

Priority: LOW

System shall auto-start on system boot (optional)

3. Security Requirements

[SEC-001]

Priority: CRITICAL

API keys and secrets must be stored in environment variables, never in code

[SEC-002]

Priority: CRITICAL

All user inputs must be validated and sanitized before processing

[SEC-003]

Priority: CRITICAL

XSS prevention: all HTML rendered in UI must be sanitized

[SEC-004]

Priority: CRITICAL

CORS must be restricted to localhost origins only

[SEC-005]

Priority: HIGH

Security headers must be set on all HTTP responses

[SEC-006]

Priority: HIGH

Rate limiting must be implemented on all API endpoints

[SEC-007]

Priority: CRITICAL

Debug mode must be disabled in production

[SEC-008]

Priority: HIGH

Dependencies must be scanned for vulnerabilities before upgrades

[SEC-009]

Priority: HIGH

Git history must be scanned for leaked secrets before commits

[SEC-010]

Priority: CRITICAL

Passwords (if implemented) must use Argon2id or bcrypt

[SEC-011]

Priority: HIGH

No sensitive data should be exposed in error messages

[SEC-012]

Priority: CRITICAL

Environment file (.env) must be in .gitignore

4. UI/UX Requirements

- [UI-001]

Priority: CRITICAL

Every authenticated screen must fit in one viewport without page-level scrolling
- [UI-002]

Priority: HIGH

Multi-step wizards required for forms exceeding viewport height
- [UI-003]

Priority: HIGH

Paginated lists and tables -- never infinite scroll without pagination
- [UI-004]

Priority: HIGH

Dark space theme with purple/indigo accent colors
- [UI-005]

Priority: HIGH

3D knowledge graph visualization using Three.js with interactive nodes
- [UI-006]

Priority: HIGH

Responsive design: desktop, tablet, and mobile viewports
- [UI-007]

Priority: CRITICAL

Chat interface with typing indicator, source citations, and message history
- [UI-008]

Priority: MEDIUM

Timeline view showing memories organized by date
- [UI-009]

Priority: MEDIUM

Dashboard with memory statistics and usage analytics
- [UI-010]

Priority: MEDIUM

Settings panel for configuring ingestion directories and model selection
- [UI-011]

Priority: CRITICAL

Landing/marketing page may scroll; authenticated pages must NOT scroll

5. Data Requirements

- [DAT-001]

Priority: CRITICAL

Vector embeddings stored in ChromaDB with persistent local storage
- [DAT-002]

Priority: HIGH

Metadata stored in SQLite for fast relational queries
- [DAT-003]

Priority: HIGH

Conversation history persisted in SQLite for multi-turn context
- [DAT-004]

Priority: HIGH

Documents must be chunked into 1500-character segments with overlap
- [DAT-005]

Priority: HIGH

Each memory must track: source, source_path, timestamp, access_count
- [DAT-006]

Priority: MEDIUM

Data directory must be configurable via environment variable
- [DAT-007]

Priority: LOW

Backup/restore of vector store and SQLite database must be supported

6. Integration Requirements

- [INT-001]

Priority: CRITICAL

Ollama integration for local LLM inference
- [INT-002]

Priority: CRITICAL

Sentence-Transformers for local embedding generation
- [INT-003]

Priority: HIGH

Chrome/Edge/Brave browser history database reading
- [INT-004]

Priority: MEDIUM

File system watching via watchfiles or polling
- [INT-005]

Priority: LOW

Future: IMAP email ingestion
- [INT-006]

Priority: LOW

Future: Slack/Discord export parsing
- [INT-007]

Priority: LOW

Future: Calendar sync (iCal/Google Calendar)

7. Performance Requirements

- [PERF-001]

Priority: CRITICAL

Query response: < 2s for 95th percentile
- [PERF-002]

Priority: HIGH

Vector search: < 200ms for 100K documents
- [PERF-003]

Priority: HIGH

Embedding: < 100ms per 100 documents
- [PERF-004]

Priority: HIGH

LLM generation: < 5s for typical responses
- [PERF-005]

Priority: MEDIUM

Ingestion: process 100+ documents/second for text files
- [PERF-006]

Priority: MEDIUM

Memory usage: < 500MB idle, < 2GB during heavy ingestion
- [PERF-007]

Priority: MEDIUM

Disk usage: < 1GB per 100K documents indexed

8. Deployment Requirements

- [DEP-001]

Priority: HIGH

Installable via pip install for Python backend
- [DEP-002]

Priority: HIGH

Frontend buildable via npm install && npm run build
- [DEP-003]

Priority: MEDIUM

Docker Compose deployment for advanced users
- [DEP-004]

Priority: HIGH

Acceptable on Windows 10+, macOS 12+, Ubuntu 20.04+
- [DEP-005]

Priority: MEDIUM

GPU acceleration optional but supported via Ollama
- [DEP-006]

Priority: CRITICAL

Documentation: README with setup, config, and usage guides
- [DEP-007]

Priority: MEDIUM

Versioned releases with changelog